10.6 The impacts of energy production

This chapter will explain:

★ the advantages and disadvantages of non-renewable sources of energy
★ the advantages and disadvantages of renewable sources of energy
★ the strategies and techniques used to increase energy supplies
★ detailed specific example: energy resource management in Sweden.

The advantages and disadvantages of non-renewable sources of energy

Coal

Coal is the dirtiest of the fossil fuels and is the major contributor to air pollution and climate change (Figure 10.32). This is the main reason why coal consumption is being phased out in so many countries. To meet the Paris Agreement on climate change (2015), coal burning for power must decline rapidly. The objective is for coal power to cease in developed countries by 2030, and in developing countries by 2050. At the same time, coal has come under increasing competitiveness from renewable energy.

▲ **Figure 10.32** Polluting brick kilns, fired by burning coal, along the bank of the Mekong River, Vietnam

The UK was the first country to develop coal-fired power. Coal consumption in the UK has fallen dramatically over the last 30 years, from 108 million tonnes in 1990 to 6.1 million tonnes in 2022. The UK's last three deep underground coal mines closed in 2015. In June 2021, the UK government confirmed its intention to phase out coal-fired electricity generation entirely by October 2024.

In 2023, global coal production reached its highest ever level. The Asia Pacific region accounted for nearly 80 per cent of global output, with production concentrated in four countries – China, Indonesia, India and Australia. China is the world's largest coal producer (56 per cent share), consumer and importer. While China has not committed to phasing out coal, it is gradually working to lower its reliance on the fuel to improve air quality and limit the impact of climate change. Closing down surplus and inefficient coal and heavy industry plants is a priority under China's 13th Five-Year Plan.

However, significant investment in new coal projects remains. China, Japan, South Korea and India together account for an estimated US$22 billion of the US$24 billion of planned investment by G20 countries in new coal projects.

Oil

Even though oil is under pressure from climate change policies and the falling cost of renewable energy, it still contributed more than 31 per cent to global primary energy consumption in 2023. Global oil production reached a record level of over 96 million barrels per day in 2023 (with consumption at over 100 million barrels a day) (Figure 10.33). Some analysts believe this might mark peak global oil consumption. However, other analysts think that peak oil will occur later in the present decade. What appears certain is that oil will remain important to global energy supply in the short and medium term, although its relative contribution will fall.

Road fuel makes up half of global oil demand. This means that the shift from petrol/diesel to electric vehicles (EVs) will have a considerable impact on the demand for oil. However, although this shift is occurring at an increasing pace in HICs, it will take place more slowly in emerging and developing countries. The rate of change will depend to a large extent on the cost of EVs. There can be little doubt that mass manufacturing will eventually make the price of EVs much more attractive. The future availability of charging stations is an important linked issue, as is the longevity of EV batteries in terms of how far an EV can be driven before the battery needs to be recharged (as well as the lifetime of the battery itself).

▲ **Figure 10.33** An oil refinery and liquefied petroleum gas tanker in Japan

Other technological advances will also limit the demand for oil. These include:

» Sustainable aviation fuel (SAF), which is aviation biofuel from sustainable sources. SAF is chemically very similar to traditional jet fuel, and its efficacy has already been proved with commercial aircraft.
» Advances in the recycling of plastics and other oil-based products. Only about 12 per cent of plastics waste is currently reused or recycled.

Natural gas

Natural gas has become an increasingly important fuel in the global energy mix in recent decades. It is the least polluting of the fossil fuels and is often viewed as a 'transition fuel' from coal and oil to renewable energy. Natural gas has displaced higher-emitting fossil fuels in a number of countries. In so doing it has improved local air quality and reduced GHG emissions. The relative contributions of coal, oil and natural gas to total global fossil fuel emissions of CO_2 in 2021 were:

» coal: 45 per cent
» oil: 35 per cent
» natural gas: 20 per cent.

Natural gas is exported across land and shallow sea borders by pipeline, and across oceans in liquefied natural gas (LNG) tankers. The role of LNG in the international trade in energy has increased significantly in recent decades.

Shale oil and gas

The exploitation of shale oil has been a relatively recent development. It has been concentrated mainly in the USA. Shale oil is extracted from oil reserves, sometimes described as 'tight oil' reserves. These are held in shales and other rock formations, from which the oil will not naturally flow freely. Shale oil has become more accessible due to advances in technology. The rapid increase in the scale of production in the USA has fundamentally changed global energy markets, because it has enabled the USA to quickly regain much of its self-sufficiency in energy. Gas can also be obtained from shale.

Although the basic technology was originally developed in the USA in the 1940s, the more recent 'shale revolution' was the result of technological advances in horizontal drilling and hydraulic fracturing (fracking). This is a controversial technique, with concerns relating to groundwater pollution and seismic problems. As a result, some countries with commercial shale deposits, such as the UK, have decided to abandon plans to commence fracking.

Nuclear power

Over the last 60 years or so, no other source of energy has created such heated discussion as nuclear power. The main concerns about nuclear power are related to:

» power plant accidents, which could release radiation into air, land and sea
» radioactive waste storage/disposal – most concern is over the small proportion of 'high-level waste'
» terrorist use of nuclear fuel for weapons
» the high construction and decommissioning costs
» the possible increase in certain types of cancer near nuclear plants.

Before the late 1970s, the rise of nuclear power seemed unstoppable. It was generally viewed as an efficient and clean source of energy. However, incidents like the serious Chernobyl disaster in Ukraine in 1986 brought any growth in the industry to a virtual halt. As nuclear technology has advanced and safety has improved, new nuclear plants have been constructed but at a limited pace. However, the Fukushima disaster in Japan in 2011 created new doubts about the safety of nuclear reactors.

The advantages of nuclear power are:

» Zero emissions of greenhouse gases and reduced reliance on imported fossil fuels.
» It is not as vulnerable to fuel price fluctuations as oil and gas.
» Uranium, the fuel for nuclear plants, is relatively plentiful.
» In recent years, nuclear plants have demonstrated a very high level of reliability in terms of energy efficiency.
» Nuclear technology has spin-offs in fields such as medicine and agriculture.

A number of countries are considering the construction of small modular reactors (SMRs). These mini-nuclear power plants generate around 450 megawatts each, about 15 per cent the amount produced by traditional nuclear plants. As they only require the space of about two football pitches, they can be constructed relatively quickly.

Activities

1 Why is coal the most important fossil fuel to phase out around the world as quickly as possible?
2 Why are so many countries still reliant on oil as a source of energy?
3 State three advantages and three disadvantages of nuclear power.

The advantages and disadvantages of renewable sources of energy

Hydroelectric power

Of the five traditional major sources of energy, hydroelectricity is the only one that is renewable. Five countries accounted for almost 58 per cent of the global total in 2023. These were:

» China: 28.9 per cent
» Brazil: 10.1 per cent
» Canada: 8.6 per cent
» The USA: 5.6 per cent
» Russia: 4.7 per cent.

Most of the best hydroelectric power (HEP) locations are already in use (Figure 10.34), so the scope for more large-scale development is limited. In many countries, though, there is scope for small-scale HEP plants to supply local communities.

Although HEP is generally seen as a 'clean' form of energy, it is not without its problems:

» Large dams and power plants can have a huge negative visual impact on the environment.
» The dams and power plants may obstruct the river for aquatic life, for example migrating fish.
» There may be a deterioration in water quality.
» A large area of land may need to be flooded to form the reservoir behind a dam.
» Submerging large forests without prior clearance can release methane, a greenhouse gas.

▲ **Figure 10.34** A hydroelectric power plant along the River Danube in Romania

Newer alternative energy sources

The first wave of interest in new alternative energy sources resulted from the energy crisis of the early 1970s. However, the relatively low price of oil from the 1980s to the opening years of the present century dampened interest in these energy sources. Growing concern about climate change and energy price rises, however, kickstarted the alternative energy industry again. Table 10.10 shows the leading countries in 2023 for renewable energy electricity generation.

▼ **Table 10.10** Renewable electricity generation by source in terawatt hours (‘other renewables’ includes geothermal, biomass, etc., but not HEP)

Country	Wind	Solar	Other renewables	Total
China	885.9	584.2	198.1	1668.2
USA	429.5	240.5	67.3	737.3
Germany	142.1	61.2	49.5	252.8
India	82.1	113.4	37.3	232.8
Brazil	95.5	51.1	55.8	202.4
Japan	10.0	97.0	42.0	149.0
UK	82.0	13.8	34.0	129.8

Wind power

In the last 10 to 15 years, wind energy has reached the ‘take-off’ stage both as a source of energy and as a manufacturing industry. In 2023, global-installed capacity reached 1017 gigawatts – 93 per cent onshore and 7 per cent offshore (Figure 10.35).

The main advantages of wind energy are that it can:

» generate significant amounts of electricity without emitting any carbon or creating polluting waste
» be harnessed to a reasonable degree in most parts of the world
» be sited in both onshore and offshore locations
» coexist with other land uses such as farming
» be cost effective – the cost per unit of electricity has fallen significantly over the last decade.

Apart from establishing new wind energy sites, **repowering** can also play an important role. This involves replacing first-generation wind turbines with modern multi-megawatt turbines, which give much better performance.

▲ **Figure 10.35** Renewable energy: a wind farm on the coast of northern Norway

On the other hand, as wind turbines have been erected in more areas of more countries, the opposition to this form of renewable energy has increased. For example:

» Some people feel that huge turbines located nearby look unattractive, disrupt beautiful scenery and have a significant impact on property values.
» There are concerns about the hum of the turbines disturbing both people and wildlife.
» Turbines can kill birds. In response, wind companies argue that they don't place turbines near migratory routes.

Solar power

The global-installed capacity of solar electricity has increased rapidly in recent years. Solar electricity is currently produced in two ways:

» Photovoltaic (PV) systems: These are solar panels that convert sunlight directly into electricity (Figure 10.36).
» Concentrating solar power (CSP) systems: These use mirrors or lenses and tracking systems to focus a large area of sunlight into a small beam. This concentrated light is then used as a heat source for a conventional thermal power plant.

In recent years, photovoltaic systems have come to dominate installed capacity. They are easier and cheaper to establish at a wide range of scales. The leading countries for installed PV systems are China, the USA, Japan, Germany, India and Italy.

▲ **Figure 10.36** Solar electricity being generated by photovoltaic panels in northern Spain

Table 10.11 outlines the advantages and disadvantages of solar power.

▼ **Table 10.11** The advantages and disadvantages of solar power

Advantages	Disadvantages

It is a completely renewable resource	The initial cost of solar farms is very high
It generates no significant noise or direct pollution	Solar power cannot be harnessed during storms, on cloudy days or at night
It requires very limited maintenance	It is of more limited use in countries with low annual hours of sunshine
Technology is improving and reducing unit costs	Large areas of land are required to capture the sun's energy in order to generate significant amounts of power
It can be used in remote areas where extending the electricity grid would be too expensive. Having a source of electricity for the first time can greatly improve people's quality of life	Considerable areas of farmland may be lost to accommodate very large solar farms, thus compromising food security
Public perception of solar power is generally positive. Solar panels are seen as less landscape intrusive than wind turbines	Sometimes the best locations for solar farms are distant from areas of high population density, resulting in high electricity transmission costs

Tidal and wave power

Tidal power plants act like underwater windmills, transforming sea currents into electrical current. Tidal power is more predictable than solar or wind power, and the infrastructure is less obtrusive. However, start-up costs are high. Thus, the 240 MW Rance Tidal Power Station in northwestern France was the only utility-scale tidal power system in the world for 45 years, until the Sihwa Lake Tidal Power Station opened in South Korea in 2011. It uses sea wall defence barriers complete with 10 turbines to generate 254 MW. Some smaller-scale systems are in operation at other locations.

Although currently in its infancy, a study by the Electric Power Research Institute estimated that as much as 10 per cent of US electricity could eventually be supplied by tidal energy. This potential could be equalled in the UK and surpassed in Canada. So for some countries, potential energy production from this source could be very high.

Wave energy is where generators are placed on the ocean's surface and energy levels are determined by the strength of the waves. The first experimental wave farm opened in Portugal in 2008 at the Aguçadoura Wave Farm. However, the facility was shut down two months after opening due to technical problems. A number of research projects are in operation, including one off the shores of Oregon in the USA. The costs and benefits of wave energy are broadly similar to those of tidal power.

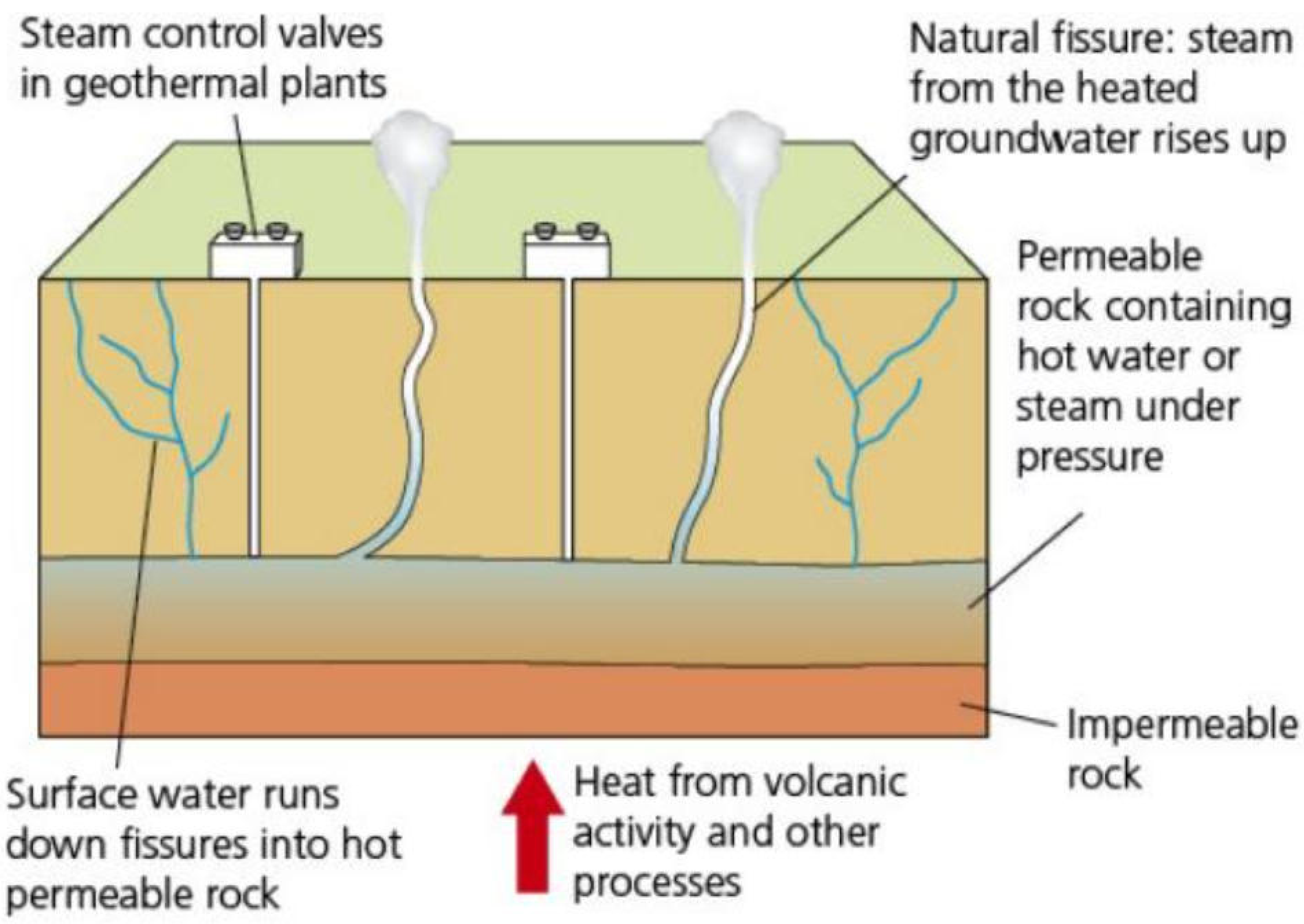
Biomass

Biomass can be used to produce renewable energy, thermal (heat) energy and transportation fuels (biofuels). Biofuels are fossil fuel substitutes that can be made from a range of crops, including oilseeds, wheat and sugar. They can be blended with petrol and diesel. The biggest producers of biofuels are the USA, Brazil, Indonesia, China and Germany. The USA and Brazil together accounted for almost 60 per cent of global biofuels production in 2023.

By increasing biofuel production, these countries have reduced the amount of oil they need to consume. This is the main reason behind biofuel production. Advocates of biofuels also argue that biofuels come from a renewable resource (crops); they can be produced wherever there is sufficient crop growth, helping energy security; and they often produce fewer emissions than petroleum-based fuels. However, there are clear disadvantages in biofuel production. Increasing amounts of cropland have been used to produce biofuels, adding to the 'global food crisis'. Large amounts of land, water and fertilisers are needed for large-scale crop production. The manufacture of biofuels also uses significant amounts of energy, creating greenhouse gas emissions.

Geothermal electricity

Geothermal energy is the natural heat found in the Earth's crust in the form of steam, hot water and hot rock (Figure 10.37). Rainwater may percolate several kilometres down in permeable rocks, where it is heated due to the Earth's geothermal gradient. This is the rate at which temperature rises as depth below the surface increases. The average rise in temperature is about 30°C per km, but the gradient can reach 80°C near plate boundaries. This source of energy can be used directly for industry, agriculture, bathing and cleansing. For example, in Iceland hot springs supply water at 86°C to 95 per cent of the buildings in and around Reykjavik.



▲ **Figure 10.37** Geothermal power

The USA is the world leader in geothermal electricity. However, total production accounts for less than 1 per cent of the electricity that the USA uses. Other leading countries using geothermal electricity are Indonesia, the Philippines, Turkey, New Zealand (Figure 10.38), Mexico, Italy and Iceland.

▲ **Figure 10.38** A geothermal power plant, Wairakei, New Zealand

Table 10.12 summarises the advantages and disadvantages of this form of electricity production.

▼ **Table 10.12** The advantages and limitations of geothermal power

Advantages	**Limitations**
As a non-carbon source of power, it has an extremely low environmental impact	There are few locations worldwide where significant amounts of energy can be generated – therefore, it is of only limited use
A geothermal plant occupies a relatively small land area (compared to a wind farm or coal-fired plant)	Some of these locations are far from where the energy could be used, that is, the areas of highest energy demand
Unlike wind or solar power, it is not dependent on weather conditions	Total global generation remains very small
Maintenance costs are relatively low	The installation costs of the plant and piping are relatively high since the required depth of piping below the surface is considerable

Hydrogen fuel cells

Investment in the commercial development of hydrogen fuel cells has risen rapidly in recent years. Hydrogen fuel cells can be used:

- to power buses and other vehicles
- to power generators on construction sites, replacing diesel generators
- for a range of other uses.

In terms of buses, it takes as little as five minutes each day to top up each hydrogen fuel cell at a fuelling station where hydrogen dispensers, about the size of a traditional petrol pump, are located.

Activities

1 Suggest why the location of hydroelectric power is so spatially concentrated both between and within countries.
2 Give two advantages and two disadvantages for each of the following forms of renewable energy:
   - Wind
   - Solar
   - Biofuels
   - Geothermal
   - Tidal.
3 For the country in which you live, find out which forms of renewable energy are used and how much they contribute to total energy production.

The strategies and techniques used to increase energy supplies

A range of strategies and techniques are being used by countries and energy companies to increase energy supply.

Improving the efficiency of existing fossil fuel power plants

The efficiency of older fossil fuel power plants, which mainly burn coal, becomes degraded over time. Lower efficiency reduces the power generated per unit of fuel input, and it also results in more CO_2 being emitted per unit of electricity generated. The options that are most often considered for increasing the efficiency of older fossil fuel power plants include:

- » equipment refurbishment
- » plant upgrades
- » improved operation and maintenance schedules.

Fossil fuel power plants can reuse wasted heat by being adapted to combined-cycle systems. Reusing heat makes the most of the fossil fuels consumed. Some power plants have been adapted to burn biomass alongside fossil fuels. This process, which makes the consumed fossil fuels last longer, is known as co-firing.

Scale and efficiency of production

Large land (and sea) areas are required for wind and solar projects of a size that can contribute a significant increase in renewable electricity. The scale of wind and solar projects has increased considerably since the late twentieth century. Scale goes hand in hand with efficiency, as investment in technological improvements seeks to achieve more energy from each solar panel and wind turbine. In general, the larger the size of renewable energy projects, the greater the **economies of scale** that can be achieved. However, lagging well behind the significant scaling up of renewal energy production is battery energy storage, which needs to be scaled up dramatically in a very short time period to have the best chance of maintaining momentum in the energy transition.

Energy intensity is a measure of energy efficiency. It is defined as consumption per unit of output. It provides a useful indication of changes in energy efficiency over time. For example, between 2010 and 2019 domestic energy consumption per household in the UK declined by 21 per cent. Reasons for this include:

- » changing to light-emitting diode (LED) and compact fluorescent light (CFL) bulbs
- » upgrading to higher energy-rated boilers and domestic appliances
- » more homes installing cavity wall and loft insulation, as well as double glazing, to reduce home heat loss
- » the rollout of smart meters.

Smart power grids are being developed in many countries to improve the efficiency of power networks. This reduces power lost by inefficient operation. This is broadly similar to reducing leakages in water pipe networks. **Smart meters** are an integral part of smart grids. These are installed in individual homes and premises. A meter is only smart if it can communicate effectively with other equipment. Smart meters:

- » enable households to monitor the energy they are using (and wasting) in real time by identifying the cost of running individual appliances and gadgets – this encourages people to adopt good energy habits
- » provide automatic meter readings and billing.

Smart meters enable the implementation of Home Energy Management Systems (HEMS) and Building Energy Management Systems (BEMS), allowing visualisation of power usage in individual homes or entire buildings.

Energy conservation means reducing the consumption of energy by using energy more efficiently. Apart from installing a smart meter, there are other energy-saving measures that can make a difference to households. These include: loft and cavity wall insulation, energy-efficient windows, draught-proofing windows and doors, top-rated energy-saving appliances, energy-efficient light bulbs, washing at cooler temperatures, switching off ‘standby’, making the most of natural light, and using a microwave instead of an oven where appropriate.

Managing energy supply is often about balancing socioeconomic and environmental needs. Many countries are looking increasingly at the concept of **community energy**. This is energy produced close to the point of consumption. Much energy is lost in transmission if the source of supply is a long way away. Energy produced locally is much more efficient. This will invariably involve **microgeneration**, which refers to generators producing electricity with an output of less than 50 KW.

Energy storage and connectivity to grids

Energy storage is not a new phenomenon, as pumped hydro-storage has been used worldwide for decades. In contrast, modern **battery energy storage systems (BESS)** enable excess energy from renewables to be stored and then released when the power is needed most. BESS makes renewable energy more reliable and therefore more investable. The supply of wind and solar energy can fluctuate. BESS are able to ‘smooth out’ this flow to provide a continuous power supply. Lithium-ion batteries are currently the dominant storage technology for large-scale BESS.

Although the use of battery energy storage in power systems is increasing, it is doing so from a low base. Only 16 GW of new storage systems were deployed in 2022, compared to 192 GW of solar and 75 GW of wind energy installed in the same year. The World Economic Forum (WEF) has stated that annual additions of grid-scale battery energy storage need to increase to an average of 120 GW annually to meet agreed climate change targets. To date, most battery storage capacity has been in advanced economies. However, the World Bank’s Energy Sector Management Assistance Program (ESMAP) is working to provide concessional financing for battery energy storage projects in developing countries.

Energy (electricity) grids are the framework for power distribution within a country or geographical area, and increasingly electricity grids link different countries. The US power grid is a vast interconnected network serving almost 400 million consumers across the continent. It is sometimes referred to as ‘the world’s largest machine’. This grid has been developed by linking five smaller interconnections.

The energy transition

The transition to renewable energy sources, coupled with economic growth, will result in an increasing demand for electricity. This is estimated to rise by 40 per cent between 2020 and 2030, and to double by 2050. Existing energy grids were designed for centralised, mainly fossil fuel generation. The best locations for wind and solar power do not always match the geographical pattern of existing grids. Also, grid infrastructure can struggle with the increased inflow of intermittent power sources (solar and wind), resulting in a growing need for complex balancing services. Many existing grids are showing signs of deterioration due to age.

Sustainable development is vital if the world is to limit the impact of climate change and to avoid the worst of so many other forms of environmental damage. 'Affordable and clean energy' is one of the UN's Sustainable Development Goals. In 2020 the UN made the following points about progress to achieving the objectives of Goal 7:

» There are encouraging signs that energy is becoming more sustainable and widely available. Access to electricity in LICs has begun to accelerate, energy efficiency continues to improve, and renewable energy is making impressive gains in the electricity sector.
» Financial flows to developing countries for renewable energy are increasing, but only 12 per cent goes to LICs.
» More effort is needed to improve access to clean and safe cooking fuels and technologies for 3 billion people, to expand the use of renewable energy beyond the electricity sector, and to increase electrification in many countries in Africa.

In December 2022, the International Monetary Fund, when discussing the enormity of the task ahead, stated that, 'the objective of this transition is not just to bring on new energy sources, but to entirely change the energy foundations of what today is a $100 trillion global economy.' Added to this is the very short timescale envisaged – little more than a quarter of a century. The International Energy Agency (IEA) sees the world moving from a fuel-intensive to a mineral-intensive energy system that will substantially increase pressure on the supply of critical minerals. The IEA estimates it takes 16 years from discovery of a mineral deposit to first production from a new mine, while some other estimates are longer.

Breaking the link between economic growth and greenhouse gas emissions will necessitate historic levels of capital expenditure in renewable energy, transport and industrial infrastructure. The predicted costs of the energy transition vary widely, with estimates ranging from US$100 trillion to US$300 trillion between 2020 and 2050. Economists say that the energy transition can be viewed as a series of shocks to the economy, both negative (such as an increase in energy costs) and positive (increased productivity created by investment in renewable technology). Economic growth could benefit significantly from the multiplier effect of this new investment.

Activities

1 How can the efficiency of existing fossil fuel power plants be improved?
2 Explain the meaning of 'energy intensity'.
3 Why are many countries investing heavily in the rollout of smart meters?
4 Why are battery energy storage systems so important in the energy transition?

Detailed specific example

Energy resource management in Sweden

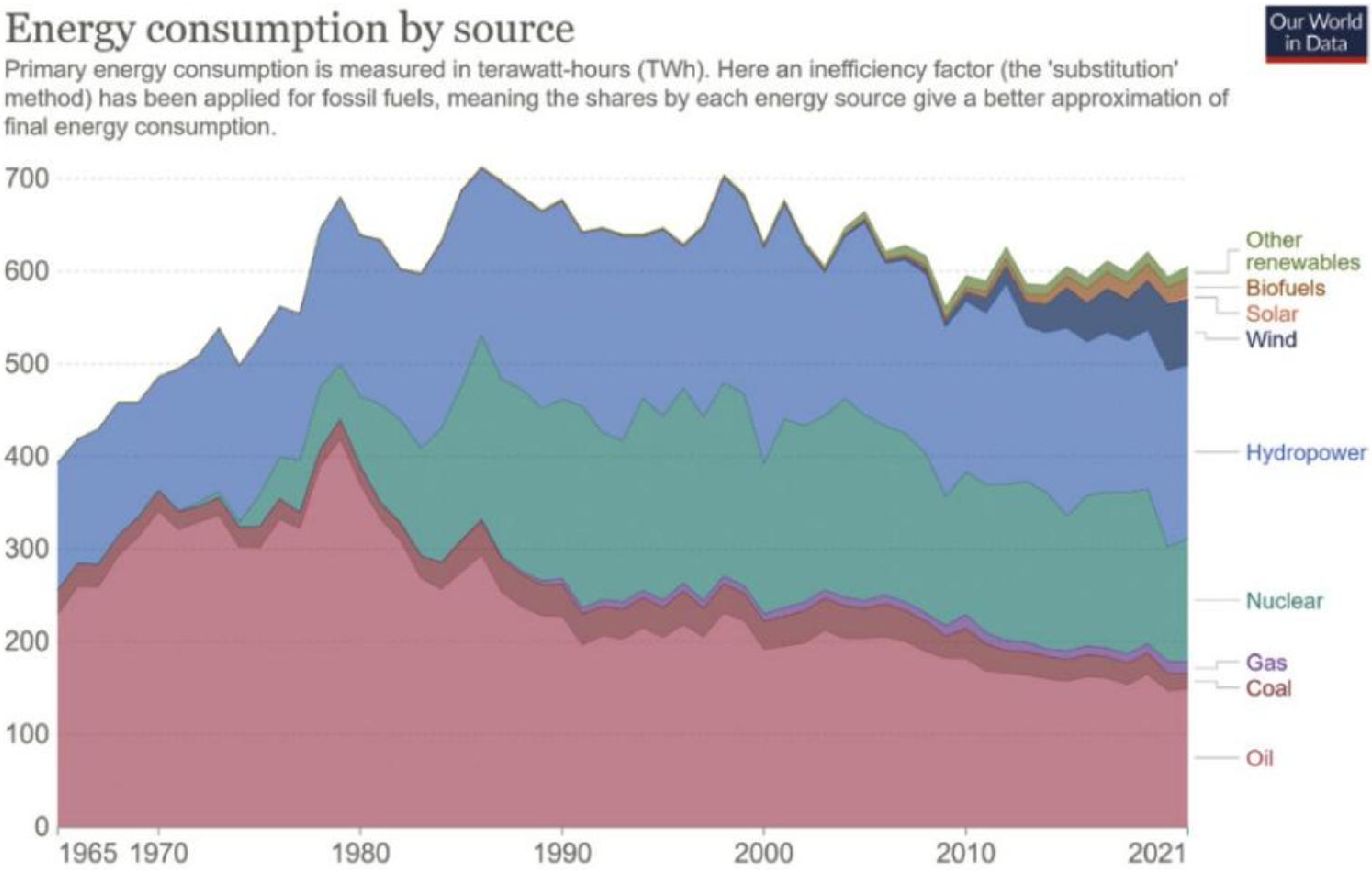
Sweden is a large, elongated Nordic country stretching from a latitude of about 55 degrees north to 69 degrees north, which is beyond the Arctic Circle. Few countries consume more energy per capita than Sweden due to a combination of its northerly location and the high average income of its population.

Sweden's energy mix

Sweden does not have any significant reserves of fossil fuels. However, its carbon emissions are very low due to a very high level of renewable energy production in the country. Sweden has the second lowest CO_2 emissions per unit of GDP and per capita among IEA members. In 2022, Sweden's primary consumption energy mix was:

- oil: 21.9 per cent
- natural gas: 1.3 per cent
- coal: 3.1 per cent
- nuclear: 20.2 per cent
- hydropower: 28.5 per cent
- other renewable energy sources: 24.6 per cent.

Electricity use in Sweden peaked in 2001 and has declined somewhat since, following the pattern of a number of other developed countries. Energy use per capita declined from 229.8 gigajoules in 2010 to 215.7 gigajoules in 2022 (Figure 10.39). This was due largely to improvements in energy efficiency, with more energy-efficient insulation, domestic appliances and lighting. The residential and services sectors use the most electricity, followed by industry and then transport. Sweden has a long-term energy policy of a controlled transition to an entirely renewable electricity system. Its overall aims are ecological sustainability, competitiveness and security of supply.



▲ **Figure 10.39** Energy consumption by source in Sweden, 1965–2021

The impacts of the different types of energy used by Sweden

Hydropower and bioenergy are the top renewable resources. Bioenergy is electricity and gas generated from organic matter (biomass). Bioenergy contributes to heating in individual homes and for district heating, as well as in electricity production and for industry. Sweden is well-endowed with high volume, fast-flowing rivers, providing ideal conditions for the production of hydroelectric power.

Forests cover 63 per cent of Sweden's land area, giving the bioenergy sector a very strong resource base. There has been a rapidly rising trend of using biofuels in the transport sector. While the majority of Sweden's electricity is produced in the north of the country, most people live in the south. Good quality and highly efficient transmission corridors are important in making the most of the country's energy resources.

Nuclear energy and hydropower account for 80 per cent of electricity production. Sweden has three nuclear power plants, with a total of eight nuclear reactors in commercial operation. However, nuclear power is a controversial subject in Sweden. About 11 per cent of electricity production is from wind power, and this is set to increase sharply in the future. Combined heat and power (CHP) plants account for 9 per cent of electrical output. These plants are mainly powered by biofuels.

Solar power remains limited in capacity largely due to Sweden's northerly latitude, but it is growing with the aid of government funding. Between 2019 and 2020 the number of grid-connected PV systems increased by 50 per cent. By the end of 2020 total solar-installed capacity had reached 1090 MW. The number of heat pumps has increased sharply since the 1990s. These pumps use renewable energy by transferring heat, mainly from the ground, which reduces demand on electricity from the grid.

Strategies to manage Sweden's energy supplies

Carbon taxation has proved effective in energy change and efficiency. Sweden's carbon tax dates from 1991 and is levied on all fossil fuels in proportion to their carbon content. Pricing carbon is a way of applying the principle of 'the polluter pays'. This helps to reduce emissions in the most cost-effective way and encourages the use of new, clean technology.

Sweden operates an electricity certificate system to support and encourage renewable electricity. Electricity retailers must buy a proportion of 'green electricity' as part of their normal supply. Electricity producers receive certification for the renewable energy they generate.

Around 500 local 'district heating' systems use excess heat to warm a large number of Swedish homes. These schemes date from the late 1940s when a power station's excess heat was first used to heat nearby buildings. Steam is forced along a network of pipes to reach people's homes.

Sweden has a high level of energy security. Apart from the factors already covered, it has electricity connections with a number of neighbouring countries, including Norway, Poland and Germany. Natural gas can be imported by pipeline from Denmark.

As a member of the EU, Sweden's energy policies stem from the broad objectives set by this multinational organisation. The long-term aims of Sweden's energy policy are:

- a transition to 100 per cent renewable electricity production by 2040
- a zero-carbon economy by 2045.

Sweden plans to achieve 50 per cent more efficient energy use by 2030. Compared to the 1980s, Sweden has 25 per cent more people and GDP has doubled, but it is now using less energy and electricity than it was then. Sweden also plans to be at the forefront of energy storage because of its large variations in seasonal demand for energy.

Activities

1 Draw a graph to show Sweden's energy mix.
2 Why are hydropower and bioenergy Sweden's top two renewable energy resources?
3 Describe Sweden's carbon taxation system.
4 What are the long-term aims of Sweden's energy policy?

Practice questions

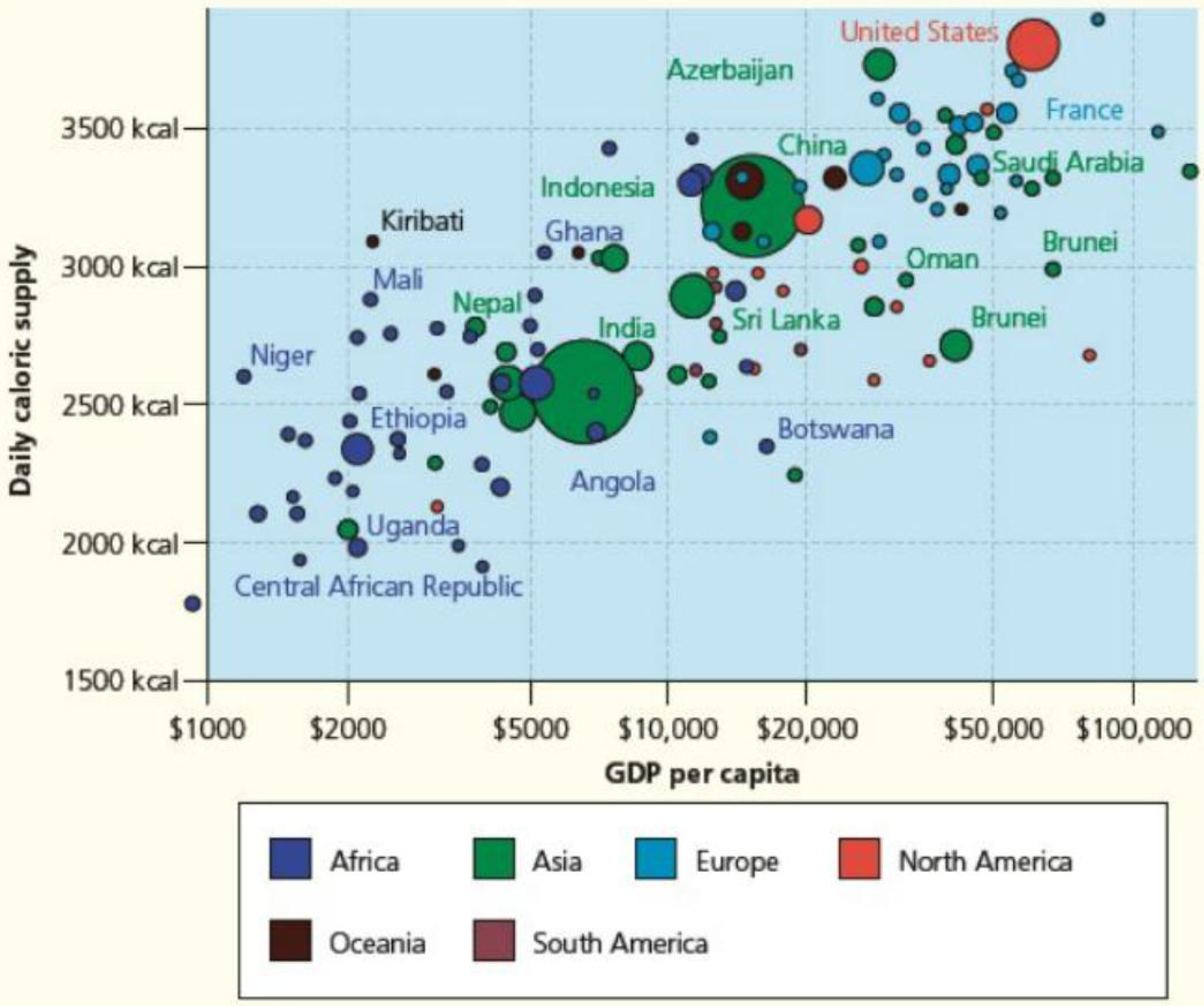
1 Figure 10.40 shows the daily supply of calories in relation to GNP.

a **i** State the approximate calorie intake for the USA and for Niger. [2]

ii Describe the relationship between calorie intake and GDP per capita as shown on the graph. [3]

b **i** Explain one human factor and one physical factor that negatively affects food supply. [4]

ii Explain two ways in which food production can be increased. [4]



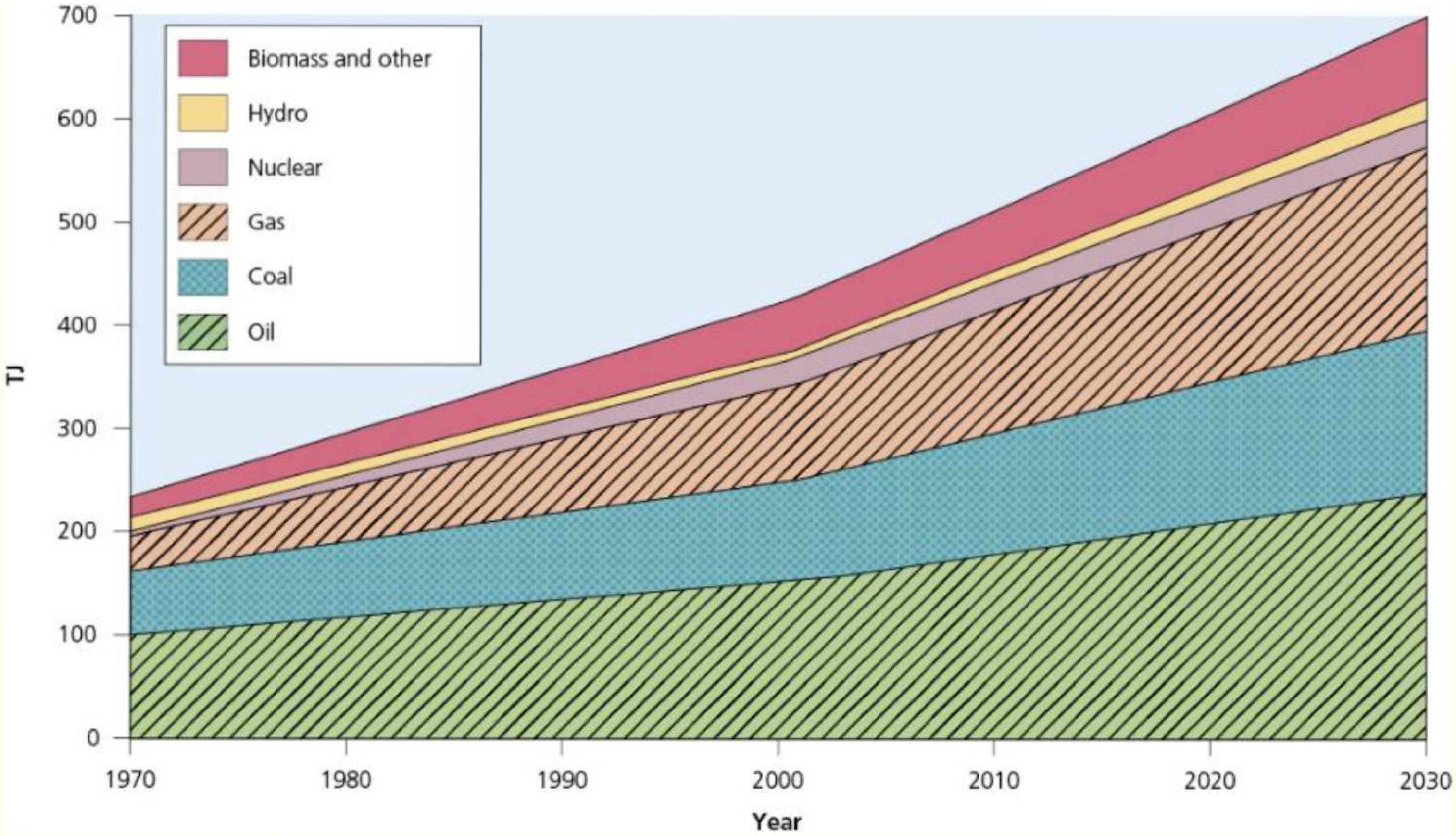
▲ **Figure 10.40** The daily supply of calories in relation to GNP

2 Figure 10.41 shows the world's different fuel sources.

a Calculate the change in the size of the world's fuel sources between 1970 and 2030 (predicted). [3]

b Identify the fuel source that is predicted to have increased most in absolute size, and the fuel source that is predicted to have increased most in relative size between 1970 and 2030. [2]

c Assess the advantages and disadvantages of different types of energy sources. [7]



▲ **Figure 10.41** The world's fuel sources, 1970–2023